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Class $\rightarrow$ KRG-1A

Expt-2

classmate

Date

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Q Algo $\rightarrow$

Set Global res = 1;

1) For each pair $[x, y]$ of two points:-
set ctr = 2;

1) Check for each point z such that $z \neq x, y$

if slope b/w $(x, z) = \text{slope b/w } (x, y)$
then $\text{ctr}++$;

(ii) set $\text{res} = \max(\text{res}, \text{ctr})$;

2) Return res;

Code :-

```
#define pii pair<int, int>
pii reducer(int x, int y, int z, int w) {
    int a = x - y, b = z - w;
    if (!a) return {x, 0}; if (!b) return {0, z};
    int g = __gcd(a, b);
    return {a/g, b/g};
}
int func(vector<vector<int>> &points) {
    int n = points.size(); res = 1 + (n > 1);
    for (int i = 0; i < n - 2; i++) {
        for (int j = i + 1; j < n - 1; j++) {
            auto [x, y] = reducer(points[i][0],
                                   points[j][0], points[i][1], points[j][1]);
            int ctr = 2;
            for (int k = j + 1; k < n; k++) {
                auto [a, b] = reducer(points[i][0],
                                       points[k][0], points[i][1], points[k][1]);
            }
        }
    }
}
```


if (a ~~==~~ x ~~&&~~ b == y) ctr++;

}

~~if~~ res = max(res, ctr);

}

}

return res;

}